

Pricing Blockchain Gossip through Bandwidth Fee Mechanisms

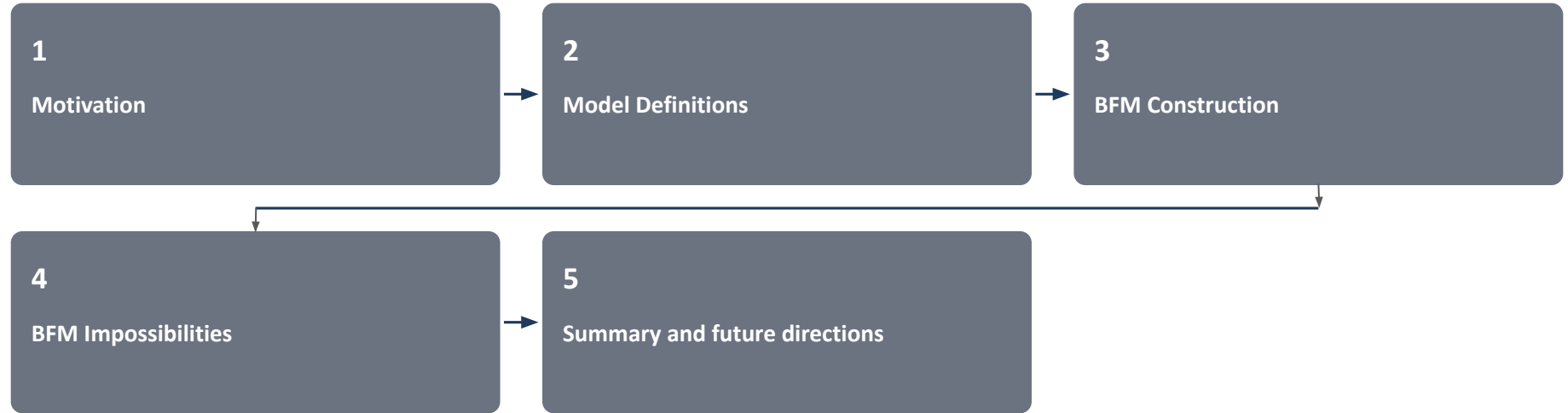
Eleftheria Fassman

In collaboration with Mahimna Kelkar (Purdue University), Benjamin Marsh (Sei Labs and University of Portsmouth), and Ke Wu (University of Michigan)

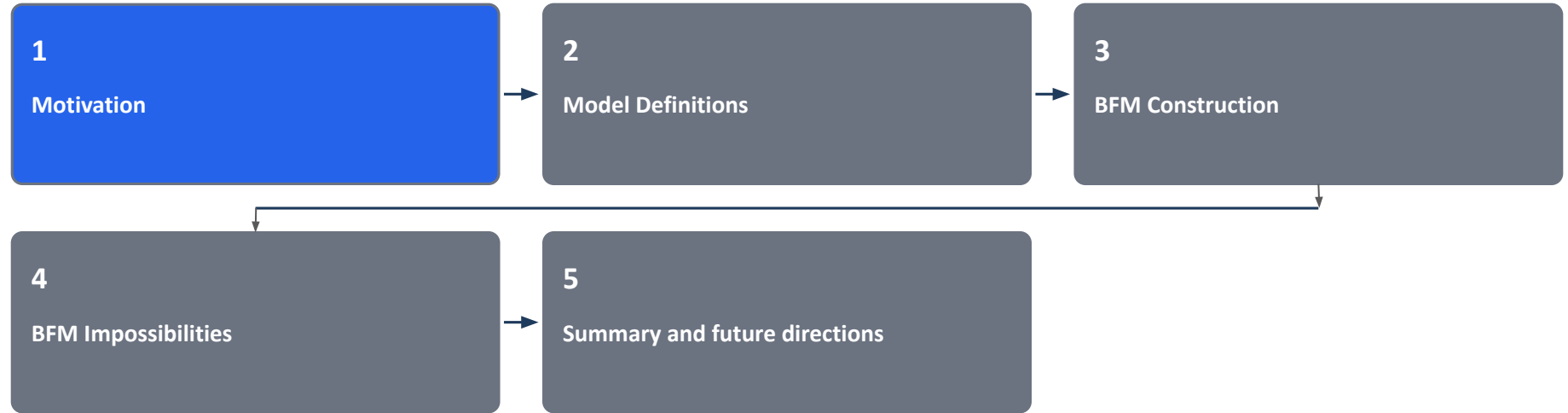
Preliminary Exam

September 11th, 2026

Roadmap

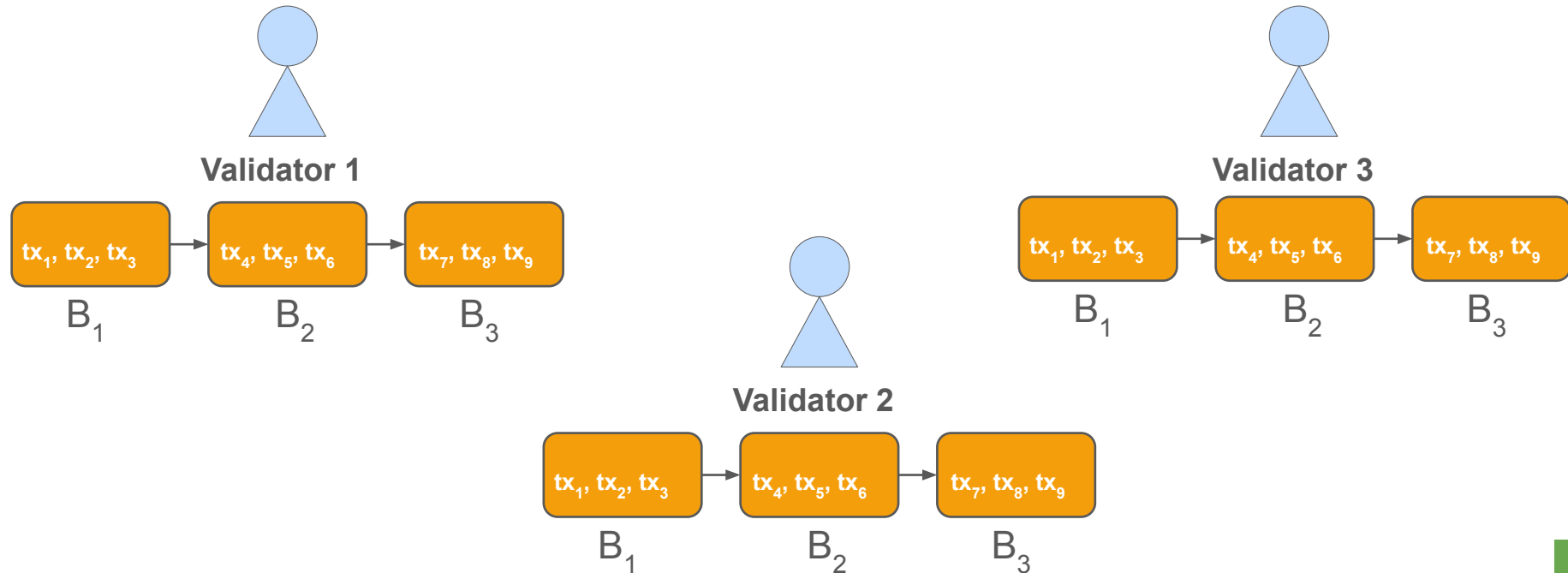


Roadmap

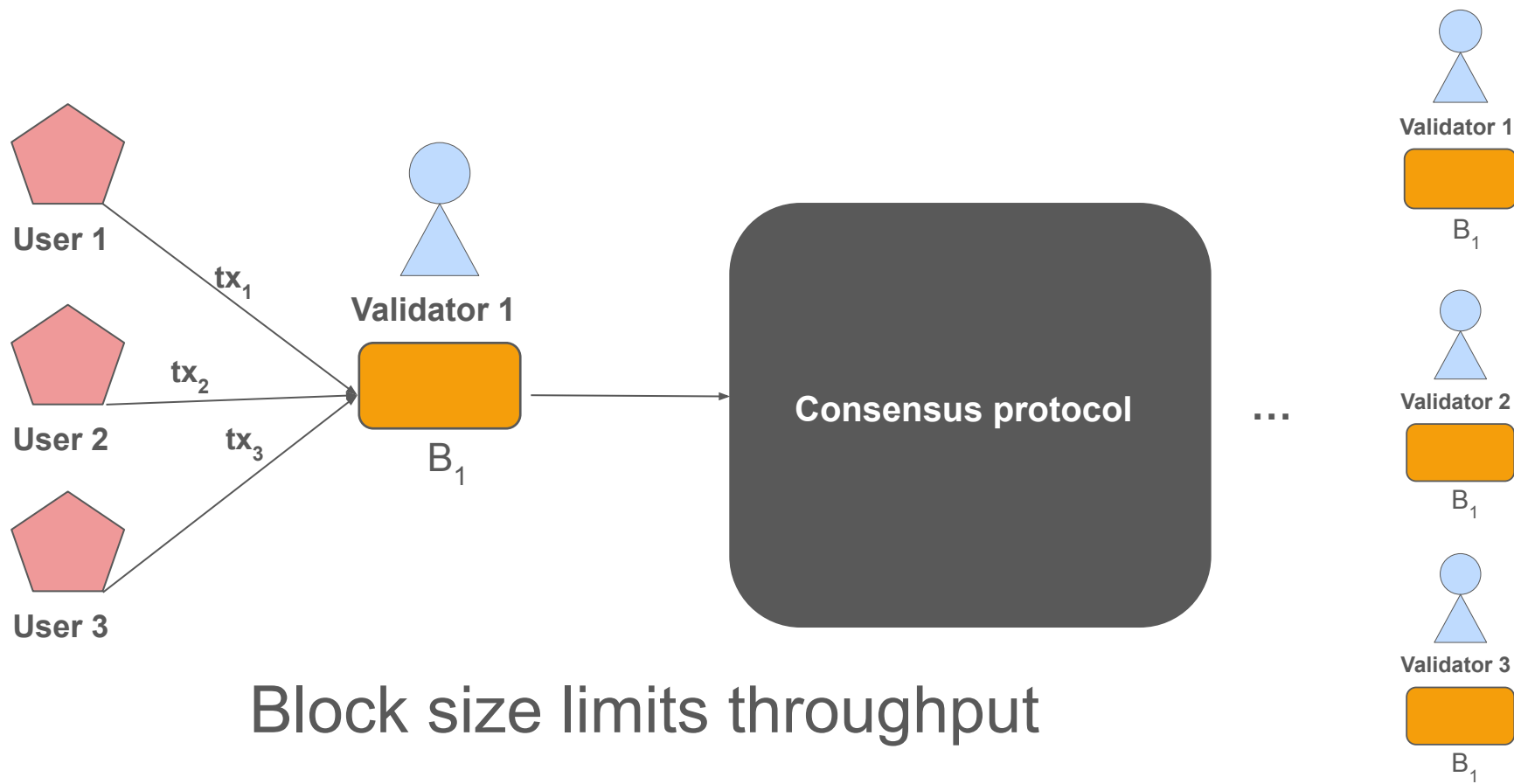


What is a Blockchain?

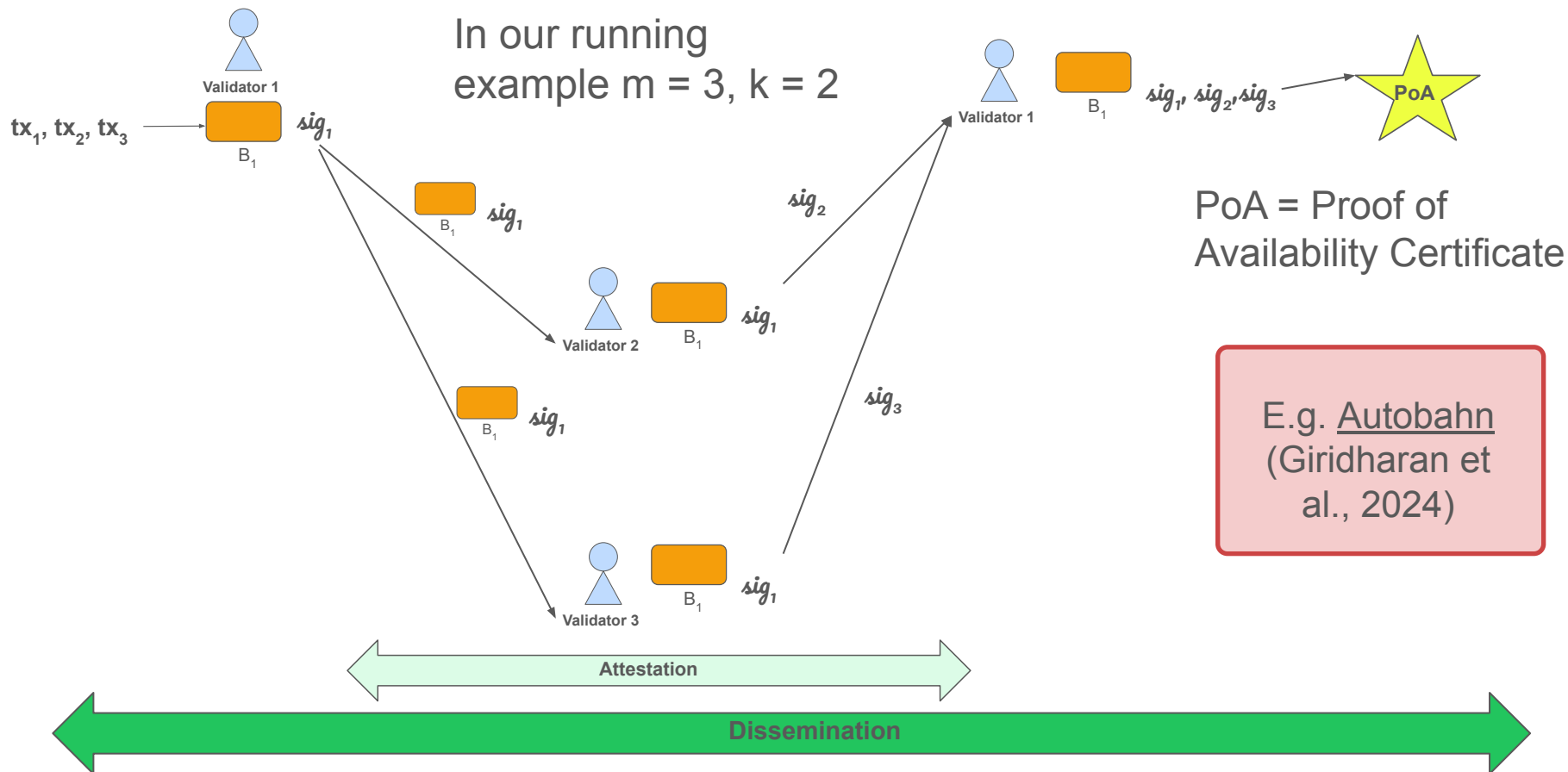
A shared, append-only distributed ledger maintained by a decentralized network



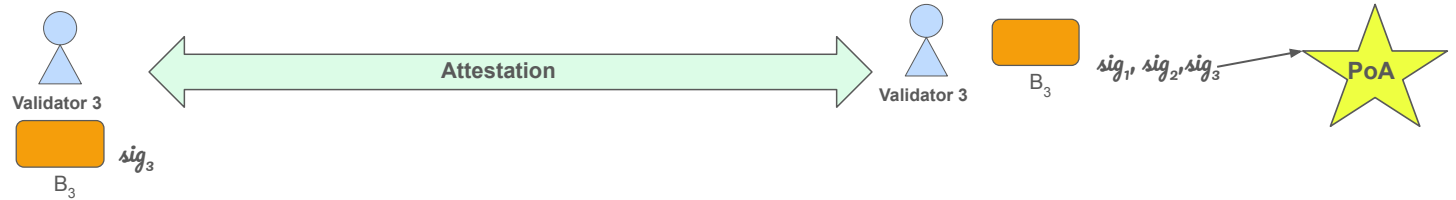
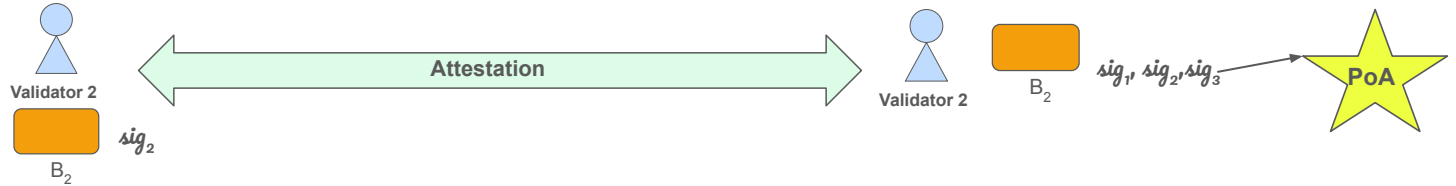
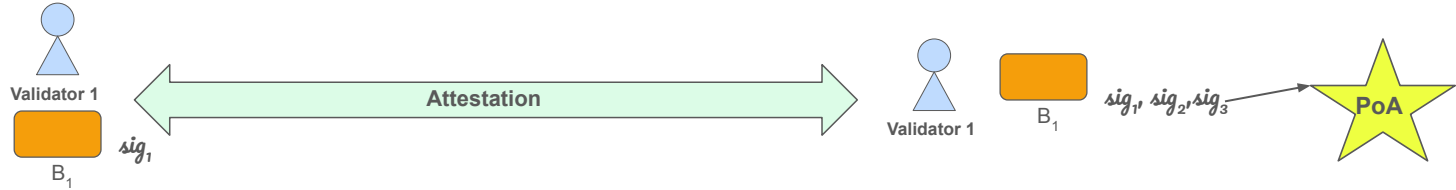
Traditional Blockchain Design



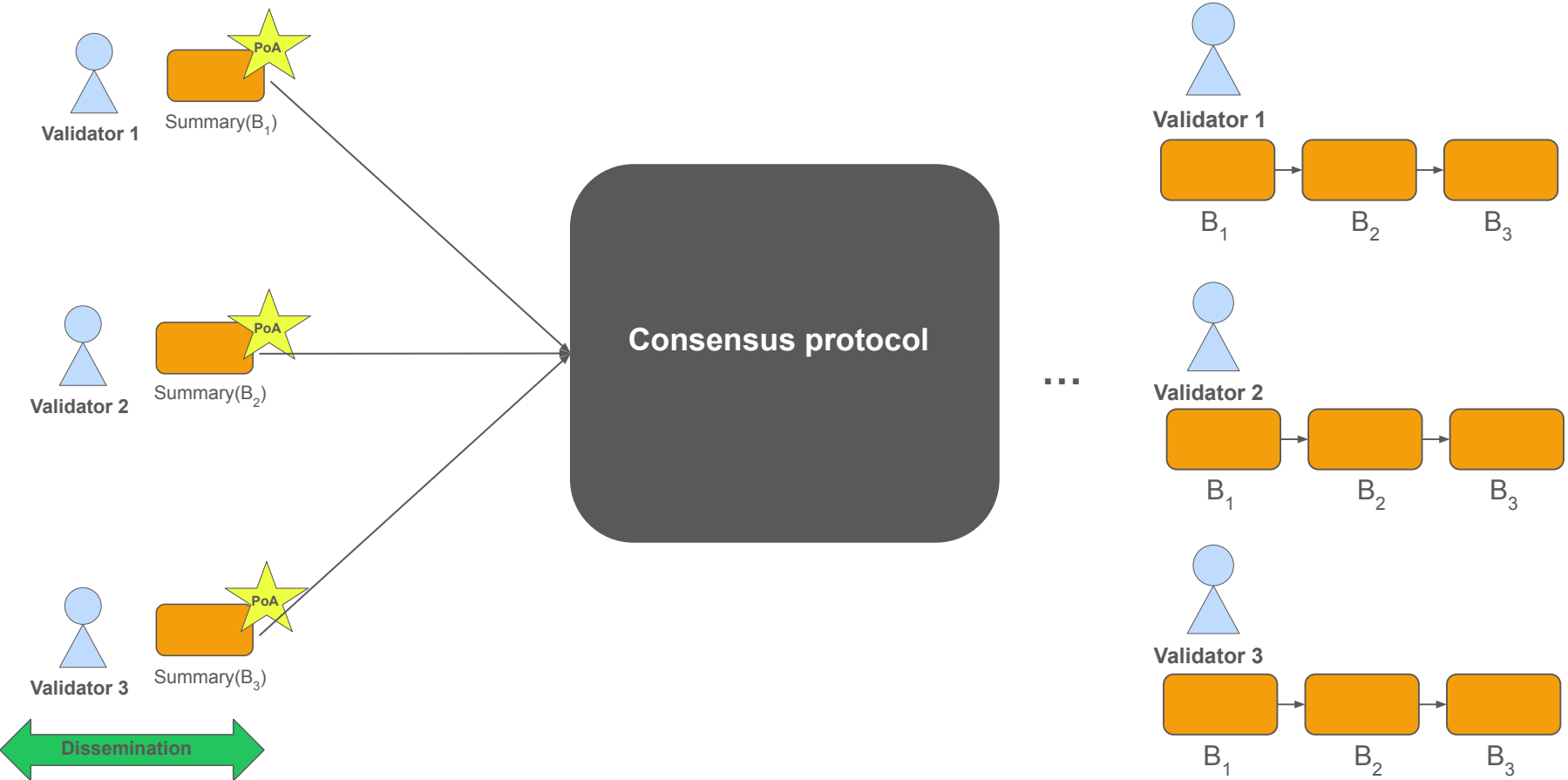
High Throughput Blockchains



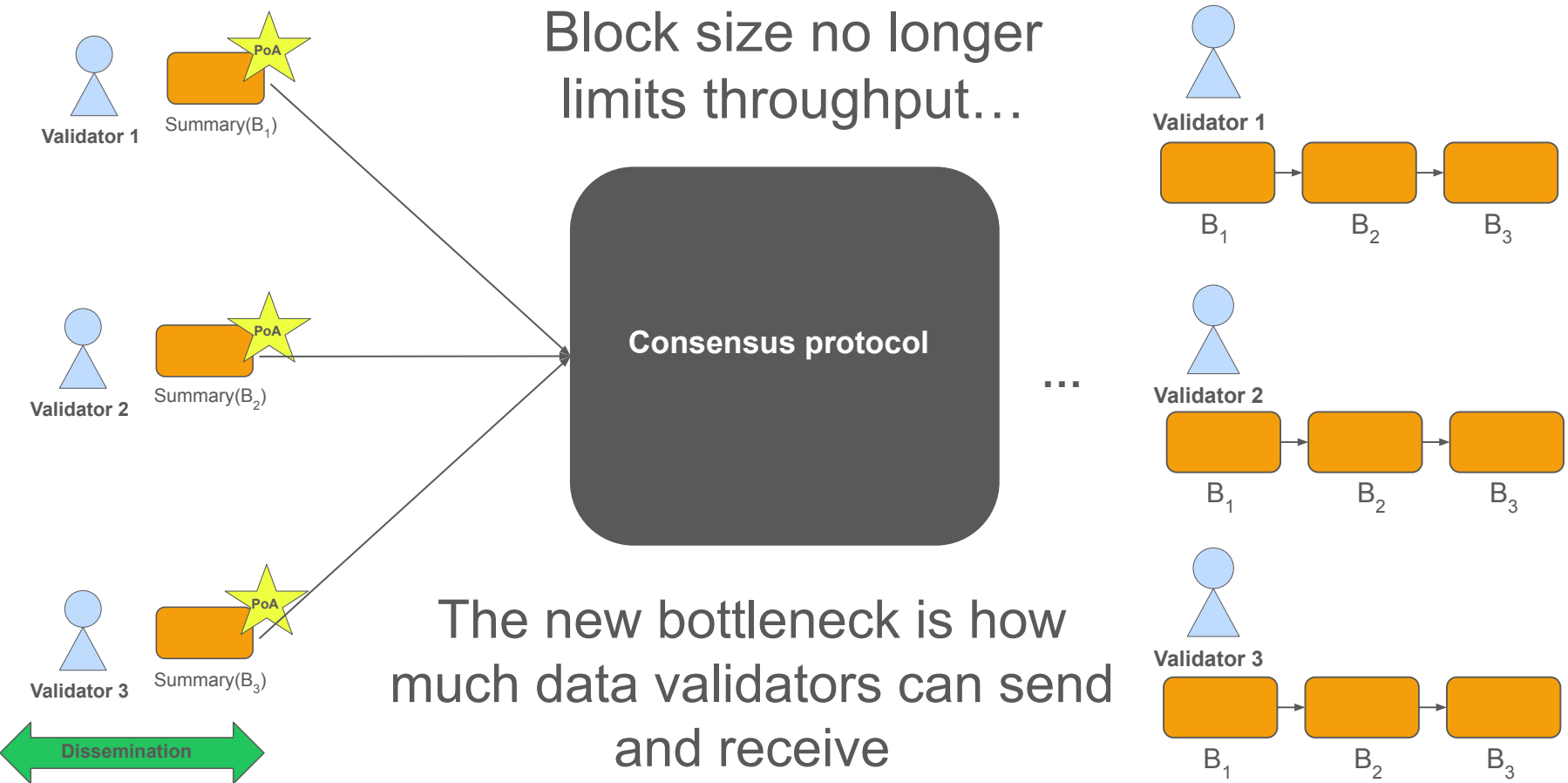
High Throughput Blockchains



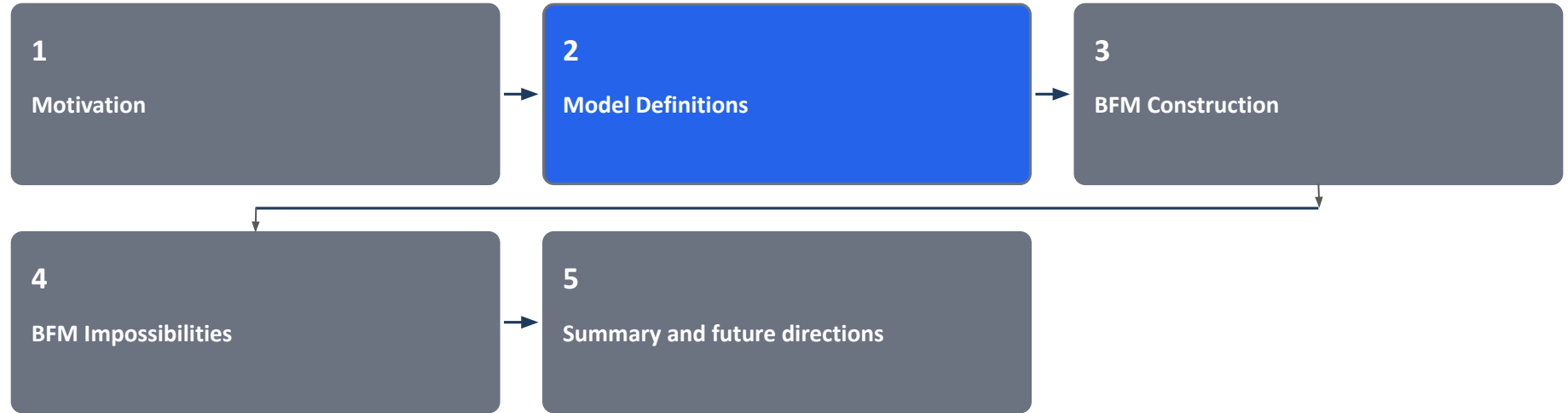
High Throughput Blockchains



High Throughput Blockchains



Roadmap



Contribution #1: Bandwidth Fee Mechanisms

A new primitive for pricing *communication* in transaction propagation

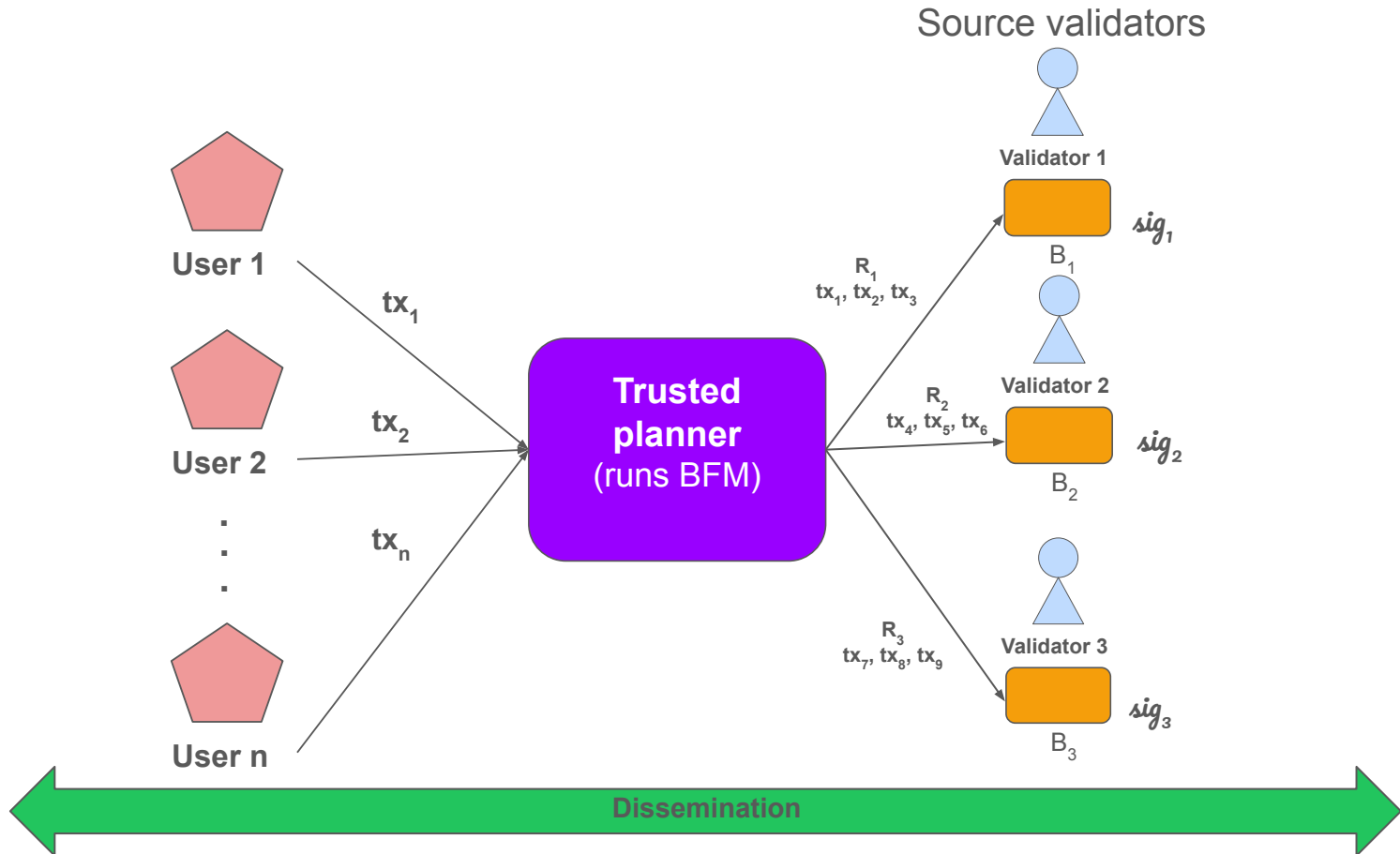
A BFM decides three things:

What transactions are handled by each validator and what attestation route are they sent along?

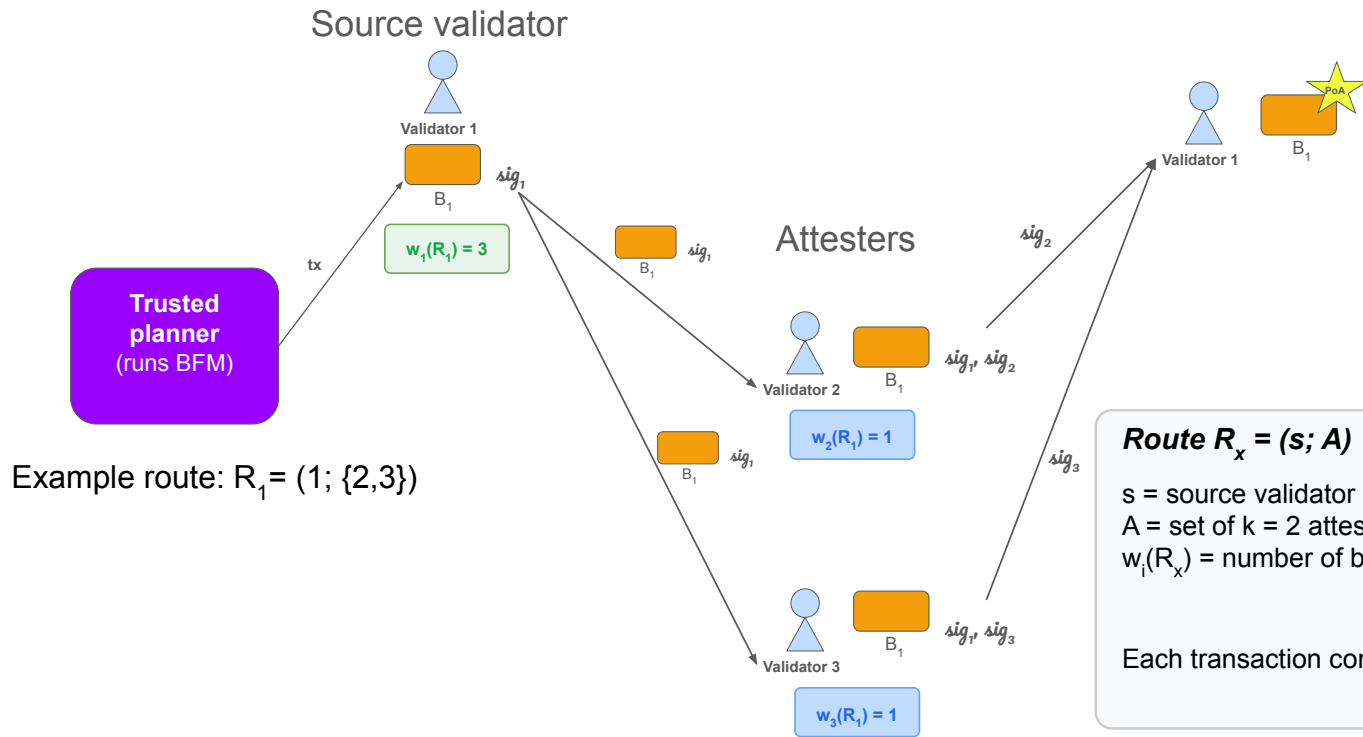
How much do users pay?

How much reward does each validator receive?

BFMs in Context



Model: How Bandwidth is Consumed



Note:

Sending/receiving signatures does not consume bandwidth (for simplicity)

Route $R_x = (s; A)$

s = source validator

A = set of $k = 2$ attesters

$w_i(R_x)$ = number of bandwidth units i spends if route R_x is used

Each transaction consumes $2k + 1 = 5$ total bandwidth units

Model: Order of the Game

1

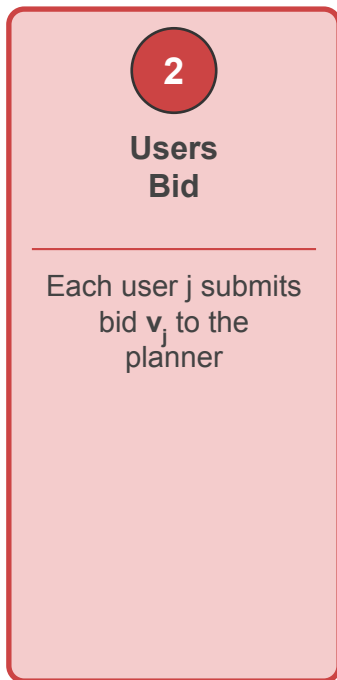
Validators Report

Each validator i
reports
cost scalar c_i and
capacity cap_i
to the planner

cap_i := the maximum number of bandwidth units validator i can spend in a single round

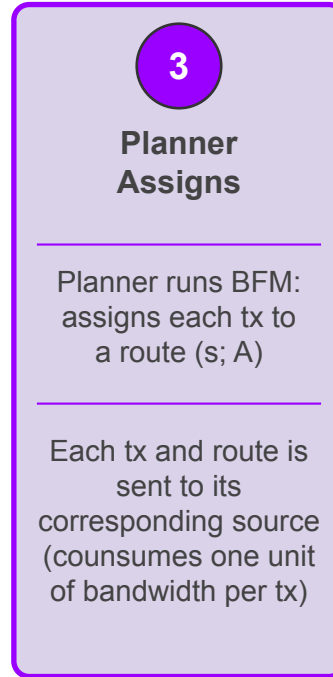
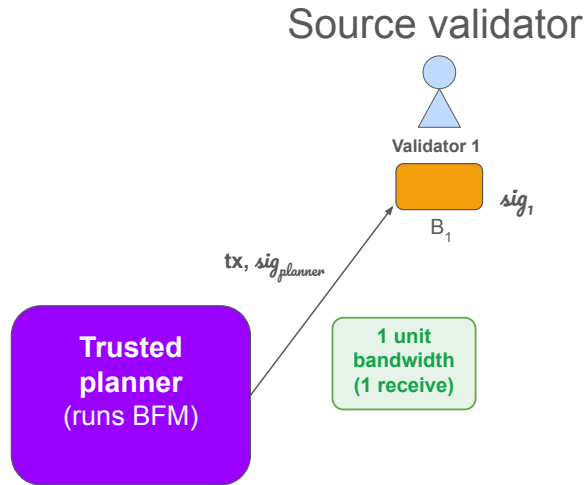
c_i := the cost scalar in the *linear* cost function: $c_i x$

Model: Order of the Game



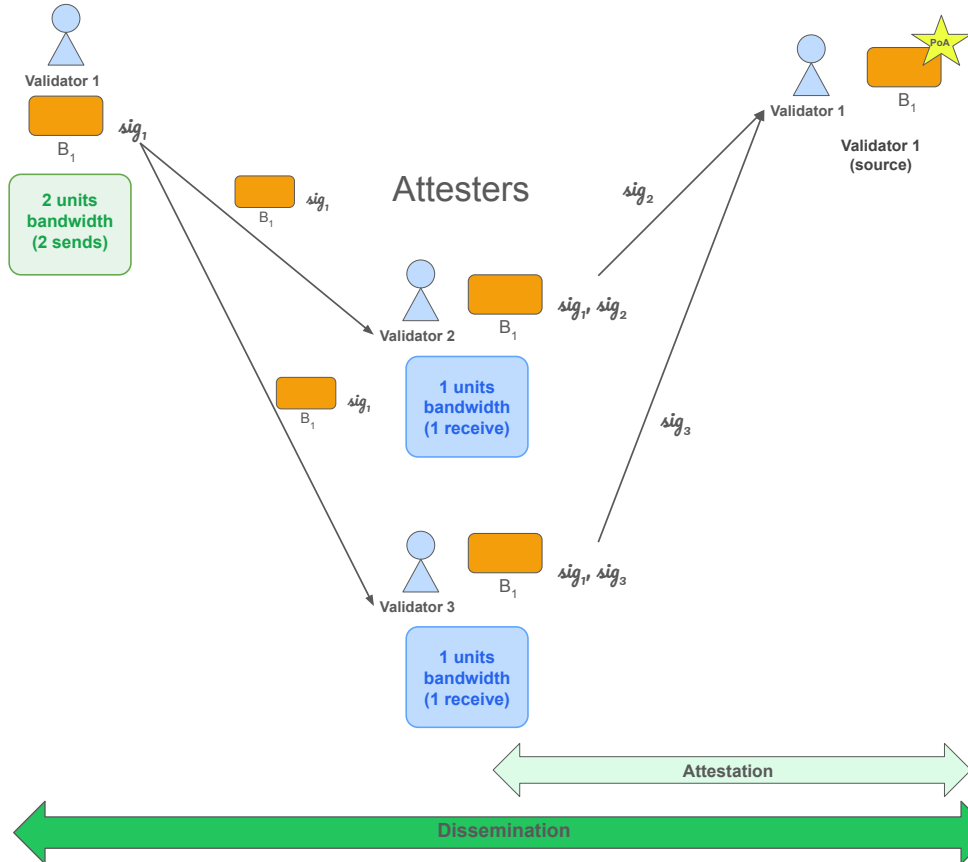
$\mathbf{v}_j$:= the maximum amount user j is willing to spend in order to have their transaction finalized on the block

Model: Order of the Game



Model: Order of the Game

Source validator



4

Dissemination & Attestation

Source forwards tx to set of attesters A, attesters then sign and return the tx to the source

Each tx consumes 2k bandwidth units during this stage

Model: Order of the Game

Payments:

Payments are determined by the payment rule of the BFM

User's payment must not exceed their submitted bid v_j

5

Payments & Rewards

Users pay if
committed to the
finalized ledger,
validators
collect rewards based
on verifiable work
appearing in the
finalized ledger

Model: Order of the Game

Rewards:

Rewards are determined by the reward rule of the BFM

Verifiable work = the number of transactions a validator sourced and attested for

Each validator's reward covers its reported cost on assigned work

5

Payments & Rewards

Users pay if committed to the finalized ledger, validators collect rewards based on verifiable work appearing in the finalized ledger

Model: Order of the Game

Budget Feasibility:

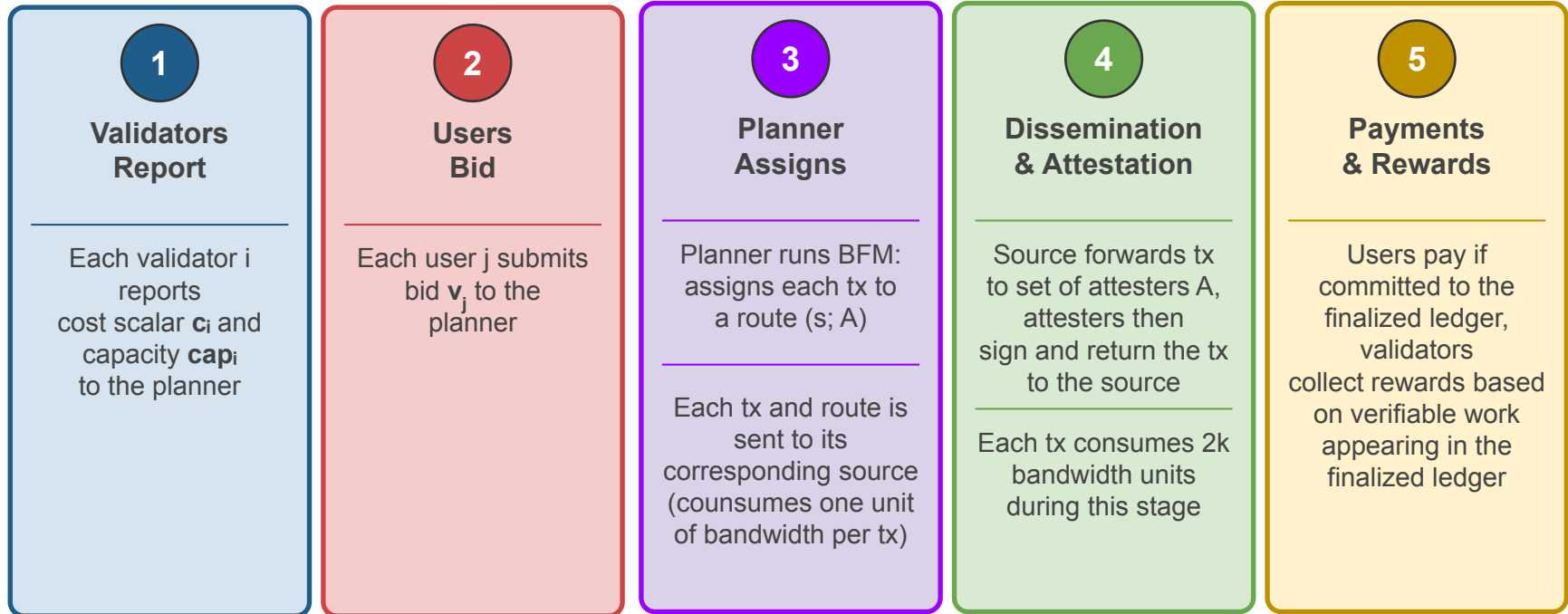
Total validator rewards $\leq$ total user payments, any surplus is burned

5

Payments & Rewards

Users pay if
committed to the
finalized ledger,
validators
collect rewards based
on verifiable work
appearing in the
finalized ledger

Model: Order of the Game



Model: Strategy Space

Honest strategy

Users

Submit a single bid equal to true valuation v^*

Strategic deviations

Submit any number of bids not equal to their true valuation ($v \neq v^*$)

Validators

Report true cost scalar c_i^* and capacity cap_i^*

As source, forward assigned transactions to correct attesters

As attester, return signatures to source

Misreport c_i or cap_i

Inject fake bids as a user

Deviate during dissemination (e.g. not forwarding tx)

Model: Utility Definitions

User utility

$$u_{\text{user}} = v_u^* \cdot \mathbb{1}_{\text{transaction finalized}} - p_u$$

- v_u^* := true private valuation of user u
- p_u := total payment charged to u
- Fake bids u injects also incur payments if finalized

Model: Utility Definitions

Validator utility

$$u_{\text{validator}} = r_i - c_i * \beta_i^{\text{act}} - p_v$$

- r_i := reward paid to validator i by the mechanism
- β_i^{act} := actual bandwidth used by i (truthfully consumed, not the reported value)
- If validator uses more than its true capacity cap_i^* (if $\beta_i^{\text{act}} > \text{cap}_i^*$), validator crashes and $u_{\text{validator}} = -\infty$

Model: Utility Definitions

User utility

$$u_{\text{user}} = v_u^* \cdot \mathbb{1}_{\text{transaction finalized}} - p_u$$

- v_u^* := true private valuation of user u
- p_u := total payment charged to u
- Fake bids u injects also incur payments if finalized

Validator utility

$$u_{\text{validator}} = r_i - c_i^* \beta_i^{\text{act}} - p_v$$

- β_i^{act} := actual bandwidth used by i (truthfully consumed, not the reported value)
- If validator uses more than its true capacity cap_i^* (if $\beta_i^{\text{act}} > \text{cap}_i^*$), validator crashes and $u_{\text{validator}} = -\infty$
- r_i := reward paid to validator i by the mechanism

Incentive Compatibility

UIC

User Incentive Compatibility

No individual user can increase its utility by deviating from honest bidding, assuming all validators behave honestly

Honest strategy

Users

Submit a single bid equal to true valuation v^*

Strategic deviations

Submit any number of bids not equal to their true valuation ($v \neq v^*$)

Incentive Compatibility

VIC

Validator Incentive Compatibility

No single validator can increase its utility by deviating from honest behavior, assuming all other users and validators behave honestly

Honest strategy

Validators

Report true cost scalar c_i^* and capacity cap_i^*

As source, forward assigned transactions to correct attesters

As attester, return signatures to source

Strategic deviations

Misreport c_i or cap_i

Inject fake bids as a user

Deviate during dissemination (e.g. not forwarding tx)

Incentive Compatibility

UIC

User Incentive Compatibility

No individual user can increase its utility by deviating from honest bidding, assuming all validators behave honestly

VIC

Validator Incentive Compatibility

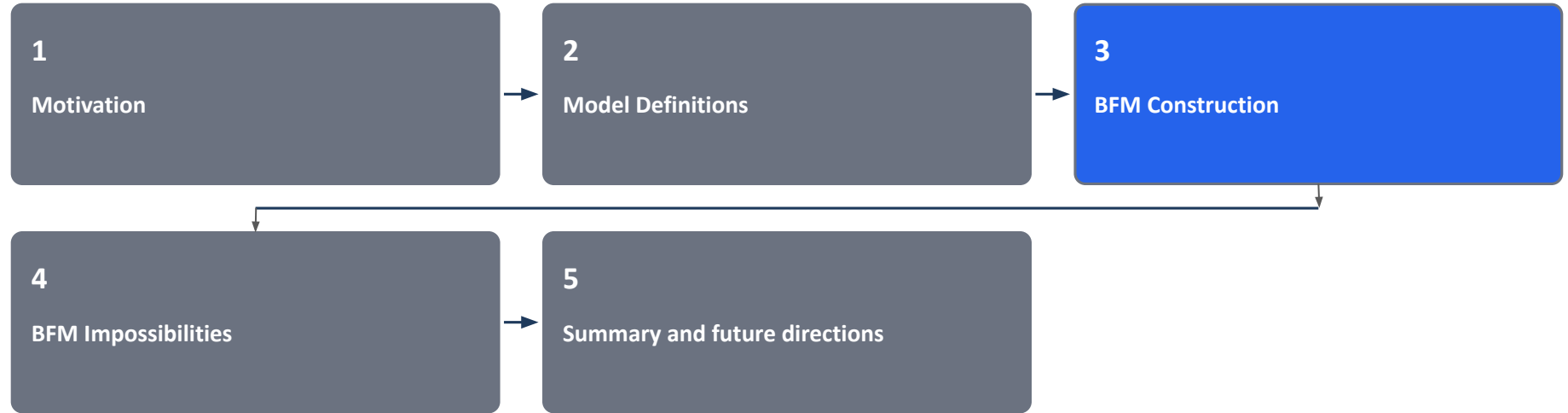
No single validator can increase its utility by deviating from honest behavior, assuming all other users and validators behave honestly

UIC and together VIC imply that
honest behavior is a Nash
equilibrium

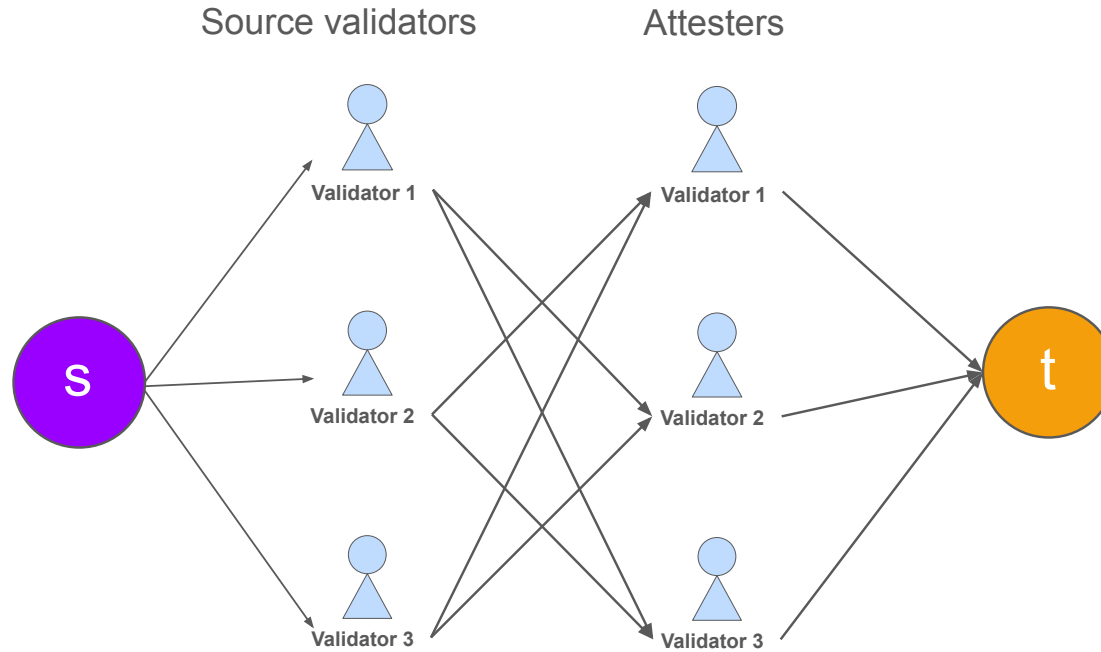


A mechanism that achieves UIC
and VIC is robust against any
single deviating party
(given honest behavior by others)

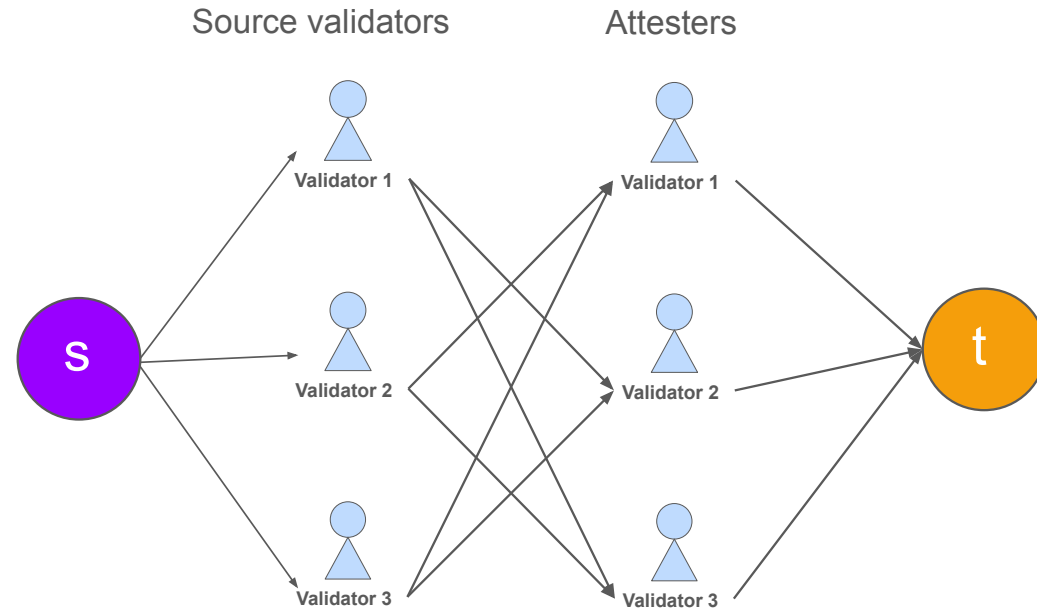
Roadmap



A Strawman Attempt: A Network Flow Approach



A Strawman Attempt: A Network Flow Approach

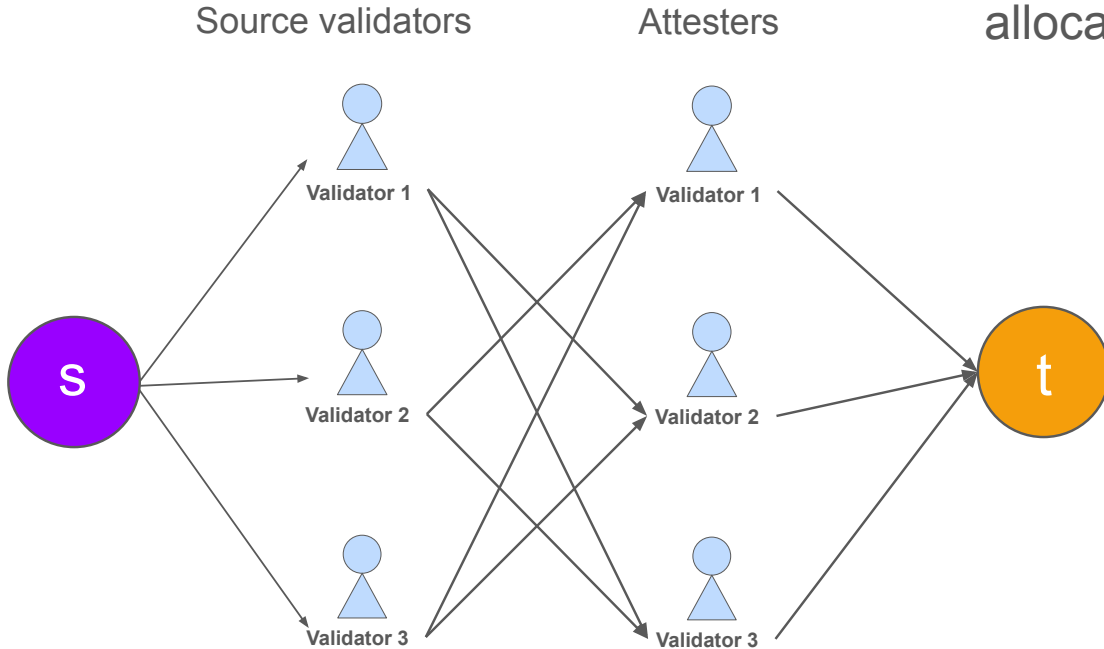


Network structure:

- 1 Linear costs $c_i x$
- 2 Each transaction consumes $2k + 1$ total bandwidth units
- 3 Fix the posted-price parameters: base fee b and reward rate τ
- 4 Transactions are the flow in the network

A Strawman Attempt: A Network Flow Approach

Solve an integer program* to find the flow allocation...



Integer program: maximize number of txs routed in the network, subject to

- 1 Capacity constraint: $\beta_i^{\text{ass}} \leq \text{cap}_i$ for all i
- 2 Budget constraint:
$$T \cdot \sum_{\text{for all } i} \beta_i^{\text{ass}} \leq b \cdot \text{total txs routed}$$

*since we measure bandwidth as integer units

The IP Fails: a Counterexample

$m = 3$ validators, $k = 2$ attesters, linear costs $f_i(x) = c_i x$, $\tau = 1$, $b = 5$

	c_i	cap_i
Validator 1	0.9	4
Validator 2	0.9	4
Validator 3	0.1	9

With 2 eligible transactions, IP must choose which 2 validators will source, lexicographic tiebreak picks R_1 and R_2

Validator 3

Consumed bandwidth β_3 : 2 txs attested = $w_3(R_1) + w_3(R_2) = 2$

Utility u_3 : $(\tau - c_3) \cdot \beta_3 = (1 - 0.1) \cdot 2 = 1.8$

Validator 3 honest utility: $u_3 = 1.8$

The IP Fails: a Counterexample

Validator 3 injects one fake eligible tx

Since $\text{cap}_1 = \text{cap}_2 = 4$, validators 1 and 2 only have capacity to attest
IP must choose R_3 for all 3 txs

Validator 3

Consumed bandwidth β_3 : 3 txs sourced = $3 \cdot w_3(R_3) = 9$

Utility u'_3 : $(\tau - c_3) \cdot \beta_3 - \mathbf{b} = (1 - 0.1) \cdot 9 - \mathbf{5} = \mathbf{3.1}$

Validator 3 honest utility: $u_3 = 1.8$

Validator 3 injection utility: $u'_3 = 3.1$

The IP Fails: a Counterexample

Validator 3 injects one fake eligible tx

Since $\text{cap}_1 = \text{cap}_2 = 4$, validators 1 and 2 only have capacity to attest
IP must choose R_3 for all 3 txs

Validator 3

Consumed bandwidth β_3 : 3 txs sourced = $3 \cdot w_3(R_3) = 9$

Utility u'_3 : $(\tau - c_3) \cdot \beta_3 - \mathbf{b} = (1 - 0.1) \cdot 9 - \mathbf{5} = \mathbf{3.1}$

Validator 3 honest utility: $u_3 = 1.8$

Validator 3 injection utility: $u'_3 = 3.1$

Fake demand triggers a global reoptimization that reshuffles the entire allocation

A Greedy Approach: The Greedy-Route Posted-Price Mechanism

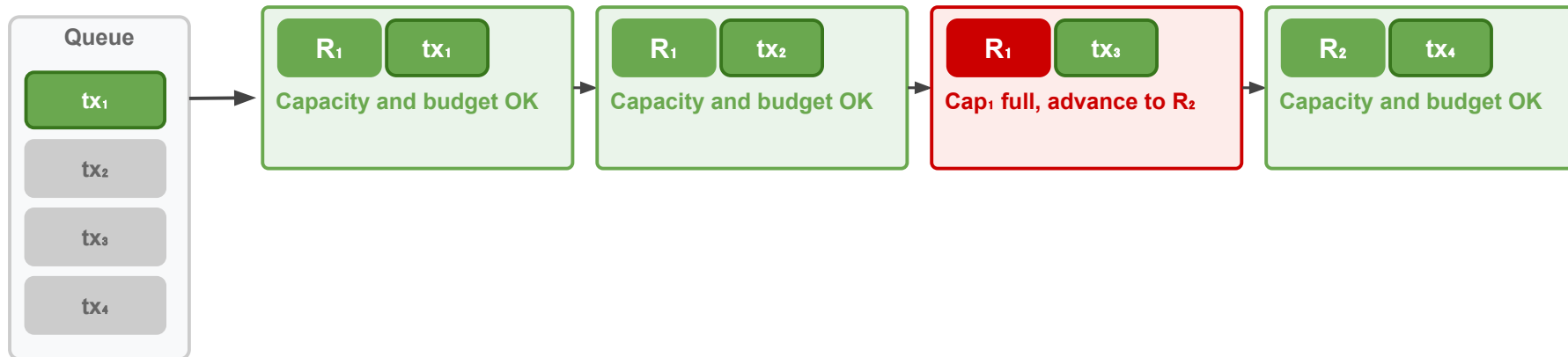
Fix base fee $b > 0$ and reward rate $0 < \tau < b$

Parameters

- Base fee b : price each user pays
- Reward rate $\tau < b$: per-unit reward to validators
- Validators with $c_i \geq \tau$ excluded

Accept route R_x if:

- (1) Capacity: $\beta_i + w_i(R_x) \leq \text{cap}_i \quad \forall i$
- (2) Budget: $\tau \cdot \sum_i w_i(R_x) \leq b$



A Greedy Approach: The Greedy-Route Posted-Price Mechanism

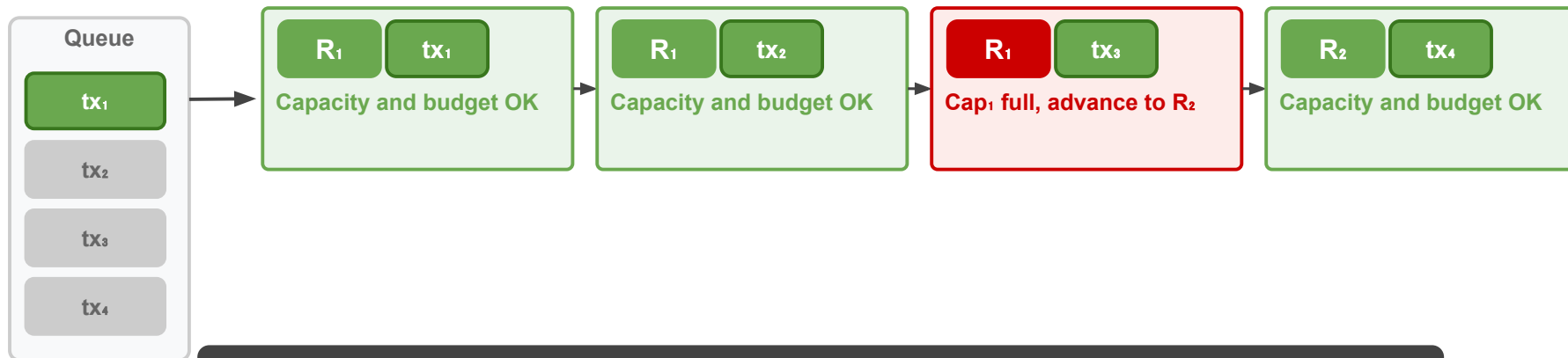
Fix base fee $b > 0$ and reward rate $0 < \tau < b$

Parameters

- Base fee b : price each user pays
- Reward rate $\tau < b$: per-unit reward to validators
- Validators with $c_i \geq \tau$ excluded

Accept route R_x if:

- (1) Capacity: $\beta_i + w_i(R_x) \leq \text{cap}_i \quad \forall i$
- (2) Budget: $\tau \cdot \sum_i w_i(R_x) \leq b$



Key: the route pointer only moves forward, no global reoptimization on new bids

A Greedy Approach: The Greedy-Route Posted-Price Mechanism

We can generalize:

c_i := the cost scalar in the cost function $f_i(x) = c_i \cdot g_i(x)$

$f_i(x)$:= captures how much it costs validator i to use x units of bandwidth

$g_i(x)$:= a publicly known shape function (need not be linear, only strictly increasing)

The Greedy-Route Posted-Price Mechanism

*The greedy mechanism extends to general m , k and strictly increasing g_i
Runs in polynomial time via DP*

Challenge: $L = m \cdot C(m-1, k)$ routes

For large m and k , explicitly enumerating all routes is exponential in k

Under general g_i , the budget check $\tau \cdot \sum_i (g_i(\beta_i + w_i(R_x)) - g_i(\beta_i))$ depends on current workloads, we cannot precompute $\delta(R_x)$ once

DP over attesters (not routes)

Fix a source s and current workloads β :

Define incremental reward for each attester j :

$$\Delta_j = \begin{cases} \tau \cdot (g_j(\beta_j + 1) - g_j(\beta_j)) & \text{if } \beta_j + 1 \leq \text{cap}_j \\ +\infty & \text{otherwise} \end{cases}$$

Finding the best k -attester set reduces to:
select k validators with minimum $\sum \Delta_j$

The Greedy-Route Posted-Price Mechanism

*The greedy mechanism extends to general m , k and strictly increasing g_i
Runs in polynomial time via DP*

Challenge: $L = m \cdot C(m-1, k)$ routes

For large m and k , explicitly enumerating all routes is exponential in k

Under general g_i , the budget check $\tau \cdot \sum_i (g_i(\beta_i + w_i(R_x)) - g_i(\beta_i))$ depends on current workloads, we cannot precompute $\delta(R_x)$ once

DP over attesters (not routes)

Fix a source s and current workloads β :

Define incremental reward for each attester j :

$$\Delta_j = \begin{cases} \tau \cdot (g_j(\beta_j + 1) - g_j(\beta_j)) & \text{if } \beta_j + 1 \leq \text{cap}_j \\ +\infty & \text{otherwise} \end{cases}$$

Finding the best k -attester set reduces to:
select k validators with minimum $\sum \Delta_j$

Runs in $O(Nm^2k)$
time

N := number
of eligible txs

Greedy vs IP Approach

The fixed route order is the source of both the mechanism's strength and its limitation

Greedy

Fake bid injection

Fixed ordering ensures a fake bid cannot change which routes get assigned to real transactions
AND
Each fake bid adds $\leq b$ reward (budget cond.) but costs exactly b , so net gain ≤ 0

IP

Global reoptimization can shift entire allocation, deviating validator gains excess reward

Greedy vs IP Approach

The fixed route order is the source of both the mechanism's strength and its limitation

Greedy

Fake bid injection

Fixed ordering ensures a fake bid cannot change which routes get assigned to real transactions
AND
Each fake bid adds $\leq b$ reward (budget cond.) but costs exactly b , so net gain ≤ 0

IP

Global reoptimization can shift entire allocation, deviating validator gains excess reward

Greedy vs IP Approach

The fixed route order is the source of both the mechanism's strength and its limitation

Greedy

IP

Cost misreporting

Threshold τ acts as posted price for validators:
 $c_i^* < \tau$ means honest inclusion is strictly better,
 $c_i^* \geq \tau$ means exclusion is correct

Same threshold applies, but global reoptimization
creates side-effects when other validators misreport

Greedy vs IP Approach

The fixed route order is the source of both the mechanism's strength and its limitation

Greedy

IP

**Capacity
misreporting**

Monotonicity: more reported capacity, weakly more assigned work, weakly higher reward

No gain from over/underreporting

Over-reporting capacity inflates workload via global reoptimization, causing a profitable deviation

Greedy vs IP Approach

The fixed route order is the source of both the mechanism's strength and its limitation

Greedy

IP

**Throughput
optimality**

Not globally optimal, fixed order may miss better allocations

(2k+1)-approximation in capacity-limited regime

Globally optimal given reported types

Greedy vs IP Approach

The fixed route order is the source of both the mechanism's strength and its limitation

	Greedy	IP
Fake bid injection	Fixed ordering ensures a fake bid cannot change which routes get assigned to real transactions AND Each fake bid adds $\leq b$ reward (budget cond.) but costs exactly b , so net gain ≤ 0	Global reoptimization can shift entire allocation, deviating validator gains excess reward
Cost misreporting	Threshold τ acts as posted price for validators: $c_i^* < \tau$ means honest inclusion is strictly better, $c_i^* \geq \tau$ means exclusion is correct	Same threshold applies, but global reoptimization creates side-effects when other validators misreport
Capacity misreporting	Monotonicity: more reported capacity, weakly more assigned work, weakly higher reward No gain from over/underreporting	Over-reporting capacity inflates workload via global reoptimization, causing a profitable deviation
Throughput optimality	Not globally optimal, fixed order may miss better allocations ($2k+1$)-approximation in capacity-limited regime	Globally optimal given reported types

Greedy vs IP Approach

The fixed route order is the source of both the mechanism's strength and its limitation

	Greedy	IP
Fake bid injection	Fixed ordering ensures a fake bid cannot change which routes get assigned to real transactions AND Each fake bid adds $\leq b$ reward (budget cond.) but costs exactly b , so net gain ≤ 0	Global reoptimization can shift entire allocation, deviating validator gains excess reward
Cost misreporting	Threshold τ acts as posted price for validators: $c_i^* < \tau$ means honest inclusion is strictly better, $c_i^* \geq \tau$ means exclusion is correct	Same threshold applies, but global reoptimization creates side-effects when other validators misreport
Capacity misreporting	Monotonicity: more reported capacity, weakly more assigned work, weakly higher reward No gain from over/underreporting	Over-reporting capacity inflates workload via global reoptimization, causing a profitable deviation
Throughput optimality	Not globally optimal, fixed order may miss better allocations ($2k+1$)-approximation in capacity-limited regime	Globally optimal given reported types

The fixed order that prevents deviation is the same property that creates the approximation gap

Recall: Incentive Compatibility

UIC

User Incentive Compatibility

No individual user can increase its utility by deviating from honest bidding, assuming all validators behave honestly

VIC

Validator Incentive Compatibility

No single validator can increase its utility by deviating from honest behavior, assuming all other users and validators behave honestly

UIC and together VIC imply that
honest behavior is a Nash
equilibrium



A mechanism that achieves UIC
and VIC is robust against any
single deviating party
(given honest behavior by others)

Incentive Compatibility of the Greedy Mechanism

Theorem (informal)

For any $m \geq 3$ and $0 < \tau < b$, the greedy-route posted-price mechanism **satisfies UIC and VIC**

This mechanism is robust against any single deviating party and honest behavior is a Nash equilibrium

Incentive Compatibility of the Greedy Mechanism

Theorem (informal)

For any $m \geq 3$ and $0 < \tau < b$, the greedy-route posted-price mechanism **satisfies UIC and VIC**

This mechanism is robust against any single deviating party and honest behavior is a Nash equilibrium

UIC

User Incentive Compatibility

Posted price: every committed tx pays exactly b , bid value is irrelevant to payment, only to eligibility

Fixed priority order: insertion-monotone rule means fake bids cannot improve real tx priority

Injection: Each fake tx committed costs exactly b with reward 0 to any one user

Incentive Compatibility of the Greedy Mechanism

Theorem (informal)

For any $m \geq 3$ and $0 < \tau < b$, the greedy-route posted-price mechanism **satisfies UIC and VIC**

This mechanism is robust against any single deviating party and honest behavior is a Nash equilibrium

VIC

Validator Incentive Compatibility

Dissemination: source only paid on completion, attestors always sign (signing is free, receiving costs bandwidth regardless)

Injection: budget condition bounds new reward per fake tx to $\leq b$, fake tx costs exactly b , so net gain ≤ 0 (holds for any g_i)

Cost/capacity misreporting: τ is a posted price threshold ($c_i^* < \tau$ means honest reporting and inclusion dominates), monotonicity rules out capacity misreports

Incentive Compatibility of the Greedy Mechanism

Theorem (informal)

For any $m \geq 3$ and $0 < \tau < b$, the greedy-route posted-price mechanism **satisfies UIC and VIC**

This mechanism is robust against any single deviating party and honest behavior is a Nash equilibrium

UIC

User Incentive Compatibility

Posted price: every committed tx pays exactly b , bid value is irrelevant to payment, only to eligibility

Fixed priority order: insertion-monotone rule means fake bids cannot improve real tx priority

Injection: Each fake tx committed costs exactly b with reward 0 to any one user

VIC

Validator Incentive Compatibility

Dissemination: source only paid on completion, attestors always sign (signing is free, receiving costs bandwidth regardless)

Injection: budget condition bounds new reward per fake tx to $\leq b$, fake tx costs exactly b , so net gain ≤ 0 (holds for any g_i)

Cost/capacity misreporting: τ is a posted price threshold ($c_i^* < \tau$ means honest reporting and inclusion dominates), monotonicity rules out capacity misreports

Incentive Compatibility of the Greedy Mechanism

Theorem (informal)

For any $m \geq 3$ and $0 < \tau < b$, the greedy-route posted-price mechanism **satisfies UIC and VIC**

This mechanism is robust against any single deviating party and honest behavior is a Nash equilibrium

UIC

User Incentive Compatibility

Posted price: every committed tx pays exactly b , bid value is irrelevant to payment, only to eligibility

Fixed priority order: insertion-monotone rule means fake bids cannot improve real tx priority

Injection: Each fake tx committed costs exactly b with reward 0 to any one user

VIC

Validator Incentive Compatibility

Dissemination: source only paid on completion, attestors always sign (signing is free, receiving costs bandwidth regardless)

Injection: budget condition bounds new reward per fake tx to $\leq b$, fake tx costs exactly b , so net gain ≤ 0 (holds for any g_i)

Cost/capacity misreporting: τ is a posted price threshold ($c_i^* < \tau$ means honest reporting and inclusion dominates), monotonicity rules out capacity misreports

Approximation Guarantee of Greedy

Theorem (informal)

If the final greedy allocation is capacity-maximal (every remaining route is blocked by some validator at capacity) then:

$$\text{Greedy} \geq \frac{1}{2k+1}OPT - \frac{(k+1)m}{2k+1}$$

Approximation Guarantee of Greedy

Theorem (informal)

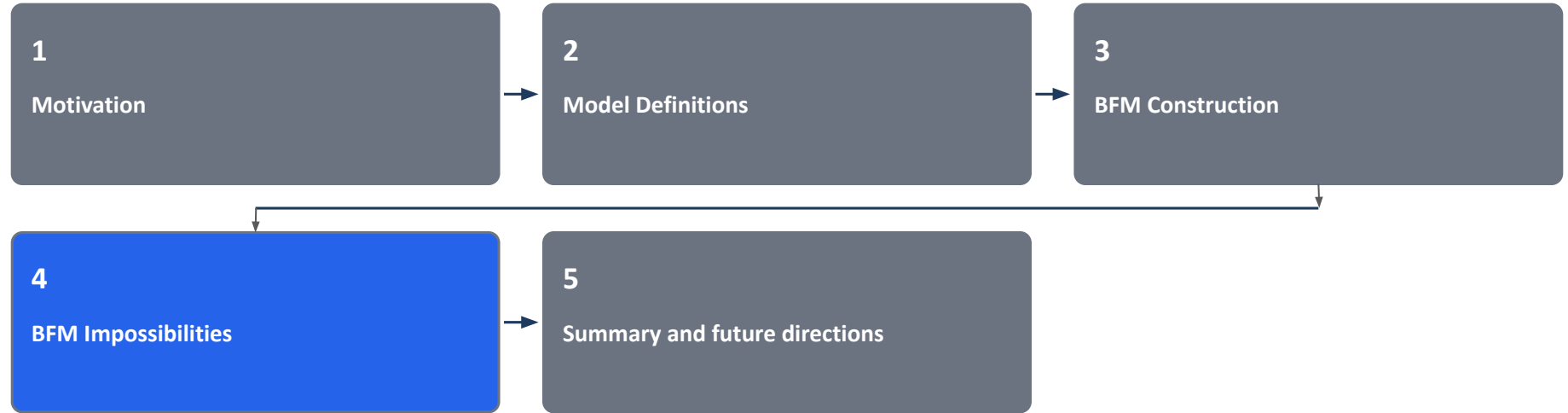
If the final greedy allocation is capacity-maximal (every remaining route is blocked by some validator at capacity) then:

$$\text{Greedy} \geq \frac{1}{2k+1} \text{OPT} - \frac{(k+1)m}{2k+1}$$

Proof intuition

- Every transaction in OPT that greedy misses must have been blocked by some validator at capacity
- Each validator can only block a bounded number of routes (each route uses $\leq k+1$ of its bandwidth)
- Charging each missed tx to its blocking validator and summing over all m validators gives the bound
- Additive slack $-\frac{(k+1)m}{2k+1}$ comes from boundary waste at each blocking validator

Roadmap



BFM Impossibilities

Two fundamental limits on what any BFM can simultaneously achieve:

Theorem: No 2-Validator Coalition Resistance (informal)

Any non-trivial BFM whose payments and rewards depend only on the public on-chain transcript *cannot prevent two validators from profitably colluding*, regardless of how the mechanism is designed

BFM Impossibilities

Two fundamental limits on what any BFM can simultaneously achieve:

Theorem: No 2-Validator Coalition Resistance (informal)

Any non-trivial BFM whose payments and rewards depend only on the public on-chain transcript *cannot prevent two validators from profitably colluding*, regardless of how the mechanism is designed

Intuition:

- Two colluding validators share signing keys, the source locally forges the attester's signature without ever transmitting the data
- This saves bandwidth cost while producing an identical on-chain transcript
- Since rewards depend only on the public transcript, the mechanism cannot detect the deviation
- This holds for any transcript-based BFM, regardless of how it is designed

BFM Impossibilities

Two fundamental limits on what any BFM can simultaneously achieve:

Theorem: No Simultaneous UIC + VIC + (1,1)-SCP (informal)

Even restricting to a single colluding validator, no non-trivial BFM can *simultaneously* protect against user manipulation (UIC), individual validator deviation (VIC), and a one-validator-one-user coalition ((1,1)-SCP)

BFM Impossibilities

Two fundamental limits on what any BFM can simultaneously achieve:

Theorem: No Simultaneous UIC + VIC + (1,1)-SCP (informal)

Even restricting to a single colluding validator, no non-trivial BFM can *simultaneously* protect against user manipulation (UIC), individual validator deviation (VIC), and a one-validator-one-user coalition ((1,1)-SCP)

Intuition:

- Via reduction to the MPC-assisted TFM impossibility (Shi *et al.*, 2023)
- Satisfying all three properties forces every validator's net utility to be exactly zero in any honest execution
- But then a validator with a lower true cost can overreport its cost scalar to extract a positive surplus, violating VIC

BFM Impossibilities

Two fundamental limits on what any BFM can simultaneously achieve:

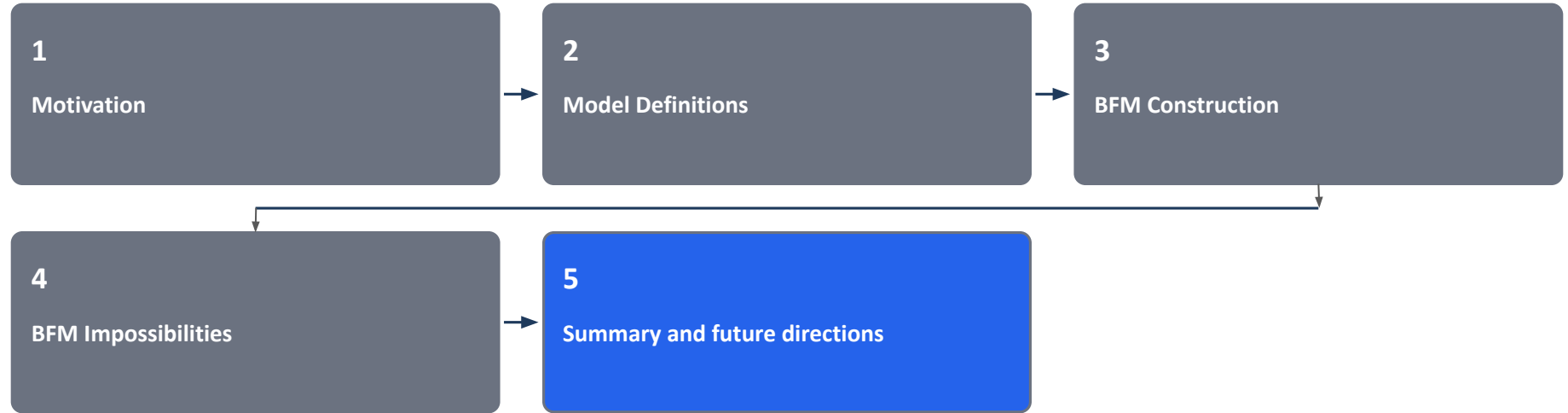
Theorem: No 2-Validator Coalition Resistance (informal)

Any non-trivial BFM whose payments and rewards depend only on the public on-chain transcript *cannot prevent two validators from profitably colluding*, regardless of how the mechanism is designed

Theorem: No Simultaneous UIC + VIC + (1,1)-SCP (informal)

Even restricting to a single colluding validator, no non-trivial BFM can *simultaneously* protect against user manipulation (UIC), individual validator deviation (VIC), and a one-validator-one-user coalition ((1,1)-SCP)

Roadmap



Summary of Contributions

1

Bandwidth Fee Mechanisms (BFMs): a New Primitive

Formalized the first
mechanism design
framework for pricing
communication in
high-throughput
blockchain
dissemination

Summary of Contributions

2

Model & Incentive Compatibility Definitions

Introduced a formal
game-theoretic model
and defined the
appropriate notions of
strategyproofness for
this setting

Summary of Contributions

3

Greedy-Route Posted-Price Mechanism

Constructed a polynomial-time BFM satisfying UIC and VIC for any $m \geq 3$

Summary of Contributions

4

Approximation Guarantee

Proved a
 $(2k+1)$ -approximation
to OPT for the greedy
mechanism

Summary of Contributions

5

Impossibility Results

Showed that preventing colluding validators is impossible and simultaneously achieving UIC, VIC, (1,1)-SCP is impossible

Summary of Contributions

1

Bandwidth Fee Mechanisms (BFMs): a New Primitive

Formalized the first mechanism design framework for pricing communication in high-throughput blockchain dissemination

2

Model & Incentive Compatibility Definitions

Introduced a formal game-theoretic model and defined the appropriate notions of strategyproofness for this setting

3

Greedy-Route Posted-Price Mechanism

Constructed a polynomial-time BFM satisfying UIC and VIC for any $m \geq 3$

4

Approximation Guarantee

Proved a $(2k+1)$ -approximation to OPT for the greedy mechanism

5

Impossibility Results

Shown that preventing colluding validators is impossible and simultaneously achieving UIC, VIC, $(1,1)$ -SCP is impossible

Future Directions

Decentralizing the Planner

Can validators self-organize route selection without a trusted coordinator?

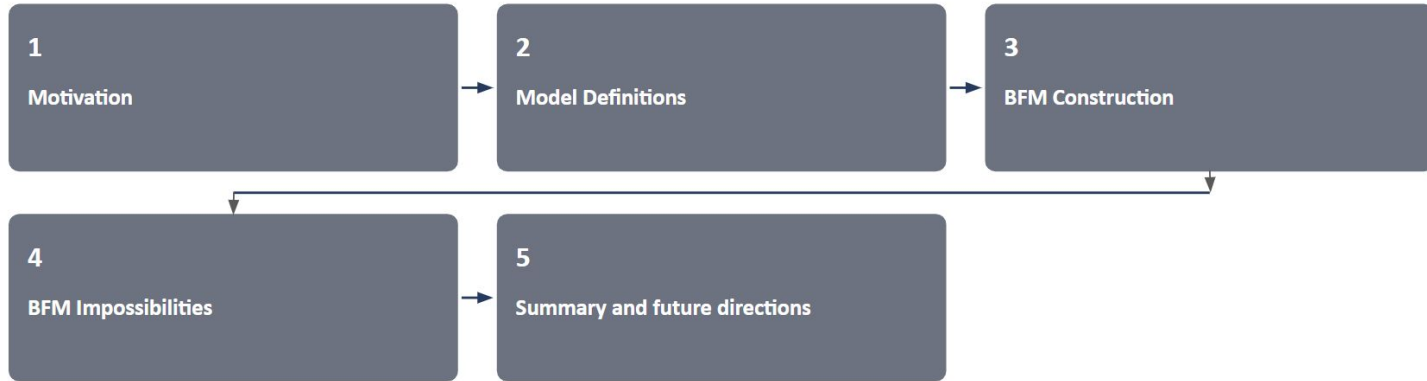
Multi-Round Strategic Behavior

Current analysis is myopic, what if validators are farsighted and strategize across blocks?

More Intelligent Transcripts

Can trusted hardware or verifiable data-possession proofs enable coalition resistance between validators (removing the key sharing attack)?

Questions?



Thank you!

